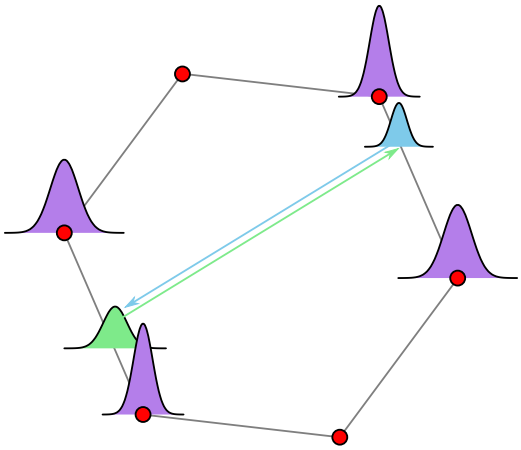
THE UNIVERSITY OF MELBOURNE

Final Report

The Art of Scientific Computing: Finite-Size Scaling of Critical Avalanches in BTW and Manna models

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Chapter 1

Introduction

The Bak-Tang-Wiesenfeld (BTW) sandpile model is a classical example that demonstrates self-organised criticality (SOC), a property of certain systems evolving towards a statistically stationary critical state without finetuning the initial conditions [1].

A system that exhibits SOC are expected to show scale invariant properties. Being scale invariant means that,

$$p(cx) = g(c)p(x)$$

where c is the scaling factor. A power law satisfies exactly this property as:

$$p(x) = Ax^{-\tau}$$

and hence,

$$p(cx) = A(cx)^{-\tau} = g(c)p(x)$$

where $g(c) = c^{-\tau}$. Thus, the distribution remains the same form under a change of scale.

Consequently, distributions of avalanches properties such as number of topplings, affected area, and radius of gyration may exhibit power-law behaviours over a truncated range, as widely studied by many research [1, 2, 3]. Notably, on a finite lattice, the power-law behaviour is truncated by a cutoff dependent on system size. The finite size scaling (FSS) hypothesis represents this systematically as

$$P(x, L) \sim x^{-\tau_x} f(\frac{x}{L^{D_x}}) \tag{1.1}$$

where τ_x is the critical exponent and the function f is the finite-size scaling function that decides the cutoff location [4].

Nevertheless, accurate estimation for these critical exponents remain elusive as research findings are not uniform. An early research by Lübeck and Usadel [2] examined the behaviour of probability distributions of topples (s), area (a) and and radius of gyration (r), and found estimates $\tau_s \approx 1.293$, $\tau_a \approx 1.330$ and $\tau_r \approx 1.665$ respectively. Meanwhile, recent work by Yadav et al. in 2022 [3] has further suggested that the area distribution contains a logarithmic dependence on system size. That is, they rescaled the probability $p(x)$ by $\log N$ where $N = L^2$ and argued the estimation should be closer to $\tau_a \approx 1$.

In addition, whether the BTW and Manna models belong to the same universality class remains a debated question. The Manna model replaces the deterministic rule in BTW with stochastic toppling [5], providing a natural extension to observe and compare their behaviours. Early work by Pietronero et al. [6] using normalisation group classified two models under the same universality class and characterised them by same exponents, with numerical estimates $\tau_a \approx 1.253$ and $\tau_a \approx 1.234$ for BTW and Manna respectively. Work by Chessa et al. [7] with data collapse and moment analysis supported this classification and estimated $\tau_a \approx 1.27$ for both models. Nevertheless, other numerical studies have reached a different conclusion. Lübeck and Usadel [2] found differences in τ_a and τ_r between the BTW and Manna models that cannot be explained by error bars. Subsequent research by Biham et al. [8] similarly argued deterministic Abelian models and stochastic models indicate different scaling behaviours and hence universality classes.

Therefore, motivated by these contrasting findings, we mainly explore two questions in this report:

1. What distributions do topples, area, loss and length exhibit, and for those that exhibit power-law behaviours with FSS, does the $\log N$ scaling proposed by Yadav et al. [3] improve data collapse?

2. How do results differ between the BTW sandpile model and Manna model?

The plan of this report is as follows. Chapter 2 discusses the fundamental properties of the BTW model, such as the role of open boundaries and the Abelian property. We also seek to highlight the difference between Manna's stochastic toppling rule and the deterministic toppling rule in BTW model. Chapter 3 explains simulations design and methods we use to measure the fits. Chapter 4 presents the numerical results of our analysis. And lastly Chapter 5 interprets our findings and reflects on limitations.

Chapter 2

Models

2.1 BTW model

In our two-dimensional BTW model, a site (x, y) becomes unstable when $z(x, y) \geq 4$. Hence an unstable site follows the deterministic toppling rule [1]:

$$z(x, y) \to z(x, y) - 4$$

And each of its closest neighbours receives one unit of sand.

$$z(x \pm 1, y) \to z(x \pm 1, y) + 1$$

$$z(x, y \pm 1) \to z(x, y \pm 1) + 1$$

BTW model is a classical example that exhibits SOC. To explain this, we follow the argument in the fundamental paper by Bak et al. [1], where they described a "minimally stable state" with all sites lie right below the toppling threshold. In our 2D case, this would mean height is 3 for all sites in the lattice. In this minimally stable state, any local perturbance will spread to its neighbours in a chain reaction and eventually propagate through the whole lattice. As the system evolves, configurations with sites further below threshold will be generated (through avalanches and dissipation) and hence break down the minimally stable clusters. This resulting critical state means that perturbations may terminate locally or propagate over a wide range of distances, so there is no characteristic length or corresponding time scale below the finite system size (hence scale invariant).

This process is also self-organised. Because when the average height of the pile is far below the threshold, most added sand units will cause little activity or loss and simply accumulates, whereas when the pile reaches close to the threshold, it becomes more susceptible. New sand additions can cause large avalanches and dissipation off the boundary and hence reduce the height. This feedback causes the system to approach a critical state without external parameter tuning.

We may later observe this in Chapter 3 Figure 3.1, where the stationary average height for the BTW model lies at around 2.1 and Manna model at around 0.7. Neither model approaches the minimally stable heights (3 and 1 respectively), which would be highly susceptible to system-wide propagation, nor does it remain at a low height where perturbations terminate almost immediately.

2.1.1 Open boundaries

In this report we conduct simulations on a finite grid with open boundaries, meaning that a sand unit will escape the grid once it reaches beyond our finite boundary. Imagine if the boundary is closed, then no sand units will be able to escape and at some point the system will be stuck in an unstable state and avalanches will never be able to terminate.

Suppose the lattice is infinite and we only drop the sand units on a finite subset. Hence no sand will escape the system and the total mass keeps increasing. The sand units will spread outwards and the area affected will be progressively larger. A paper by Ostojic [9] specifically studied this case, where sand units were added only on one site in an infinite lattice. The resulting rescaled configurations may approach a limiting spatial pattern and display self-similarity. However, the author found that the perturbed area takes different shapes with different uniform initial conditions. This behaviour is therefore not equivalent to the SOC property examined in

our finite model.

2.1.2 Stationarity

In this report, we analyse observations after the system has reached a steady and statistically stationary state. A steady state generally means the overall behaviour of a system remain constant with zero net flow. A statistically stationary state means the probability distribution is time independent. An equilibrium state means there is no sustained external input or output.

After the initial transient state, our BTW model reaches a steady and statistically stationary state rather than an equilibrium state. This is because individual configurations and avalanche sizes continue to change, but quantities such as the average pile height fluctuate around constant long run values. The model is not in equilibrium because grains are continuously added and dissipated through the open boundaries.

2.1.3 Abelian property

The topplings in the BTW sandpile model are Abelian, meaning that when there are multiple unstable sites in the system, it does not matter which order we perform each toppling. The toppling matrix $\Delta(x, y)$ redistributes sand from unstable site x to site y must therefore satisfy:

$$\forall x, y \in L, x \neq y, \Delta(x, y) = \Delta(y, x) \geq 0 \tag{2.1}$$

$$\forall x \in L, \Delta(x, x) < 0 \tag{2.2}$$

$$\forall x \in L, \sum_{y \in L} \Delta(x, y) \leq 0 \tag{2.3}$$

$$\sum_{x,y \in L} \Delta(x, y) < 0 \tag{2.4}$$

where L is the lattice of the sand pile.

Eq. (2.1) indicates that the interaction between two sites is symmetric. The sand units redistributed from x to y when x topples is equal to the units redistributed from y to x when y topples. And this redistribution amount will always be non-negative when x and y are different sites. Eq. (2.2) describes that a toppling site will lose sand units and hence the change is negative. Eq. (2.3) describes a single toppling caused by an arbitrary unstable site x cannot increase the overall mass. The total change of sand units will either equal to 0 when no sand is lost from our system (that the toppling does not reach the boundary), or be negative if some sand units are dissipated from the system. In eq. (2.4), we extend the previous case by summing over all sites in the lattice, that is toppling every site in the lattice. By the previous case the sum is automatically non-positive, and because the lattice will always contain at least one site that is at the boundary, there must be dissipation from the system, and hence the sum is strictly negative.

Specifically, the toppling matrix for our BTW model is given by:

$$\Delta(x, y) = \begin{cases} -2d & x = y \\ 1 & \text{x,y are nearest neighbours} \\ 0 & \text{Otherwise} \end{cases} \tag{2.5}$$

where $d = 2$ in our 2D case.

The stability threshold is therefore $k = 4$ and it satisfies all conditions we just described. Eq. (2.1) is satisfied since when x and y are different sites $\Delta(x, y) = \Delta(y, x)$ will either be 1 if x and y are nearest neighbours or 0 otherwise. Eq. (2.2) is satisfied since $\Delta(x, x) = -2d < 0$. Eqs. (2.3) and (2.4) are satisfied for the same reason as previously discussed.

And hence the toppling operator T_x that maps from a configuration p to p' is:

$$(T_x p)(y) := \begin{cases} p(y) + \Delta(x, y) & p(x) \geq -\Delta(x, x) \\ p(y) & \text{Otherwise} \end{cases} \tag{2.6}$$

Suppose $x \neq x'$ are both unstable,

$$(T_x T_{x'} p)(y) = (T_{x'} p)(y) + \Delta(x, y) = p(y) + \Delta(x', y) + \Delta(x, y)$$

$$(T_{x'} T_x p)(y) = (T_x p)(y) + \Delta(x', y) = p(y) + \Delta(x, y) + \Delta(x', y)$$

Therefore, for unstable configurations we have $(T_x T_{x'} p)(y) = (T_{x'} T_x p)(y)$.

In addition to the commutative property, we may further prove the stabilisation is unique. That is, no matter what order we take to perform the topplings through an avalanche, the result is always the same. Following Dhar's argument [10], we consider two unstable sites x and x'. By eq. (2.1), toppling one site cannot make the other stable. Repeatedly applying the property $T_x T_{x'} p = T_{x'} T_x p$ shows that changing the order of toppling a sequence of unstable sites produces the same stable configuration. Namely, if we have

$$p' = \mathbb{T} p = \prod_{i=1}^{N} T_{x_i} p$$

where $\mathbb{T}$ is the stabilisation operator through an avalanche for some unstable configuration p, p' must be unique.

2.2 Manna model

Unlike the BTW model, the Manna model has a stochastic toppling rule, so an individual toppling cannot be represented by one fixed toppling matrix. Notably, Dhar [11] established a stochastic analogue of the Abelian property for the Manna model. Nevertheless, since our report mainly focuses on the effect of stochastic toppling on avalanche statistics, we shall not discuss this property in detail.

In Manna's two state model [5], a site becomes unstable once its height reaches or exceeds 2. Hence when it topples, we have:

$$z(x, y) \to z(x, y) - 2$$

The sand units are then independently assigned to randomly selected nearest neighbours. Thus the neighbours are sampled with replacements, and it may be possible that both sand units are assigned to the same neighbour. Similar to the BTW model, any sand units directed to the outside the open boundary is lost from the system.

Chapter 3

Methodology

3.1 Simulation setup

Both models were simulated on an $L \times L$ square lattice with open boundaries. At each time step, one grain was added to a uniformly selected site using the `drop_sand` function, after which the system was relaxed completely (no unstable sites) before the next grain was added. We follow the toppling rules discussed in the previous Chapter, and any sand units directed outside the open boundary is lost from the system.

Simulations were performed for $L \in \{10, 20, 30, 40, 50\}$. This allows us to investigate the effect of system size on distributions of avalanche properties, while keeping the runtime manageable for easy replicability. Although larger lattices would provide stronger evidence especially concerning the FSS behaviour, the Manna simulations became substantially more computationally expensive as L increased. The upper system size was therefore set to $L = 50$.

In our square lattice case, the mean pile height at time t is defined as

$$< p_t >:= \frac{1}{L^2} \sum_{i,j} p_t(x, y).$$

This quantity illustrates the initial transient period before the system reaches its statistically stationary state. As shown in Figure 3.1, the BTW average height appears to stabilise after approximately 6000-7000 sand drops, whereas the Manna model stabilises earlier.

This result actually highlights the necessity of discarding the initial transient state as burn-in values. This is because sampling one system repeatedly over time is not generally equivalent to sampling an ensemble of independent systems. Only under stationary and ergodic dynamics, sufficiently long time averages then converge to its expectation under the stationary distribution π [12].

$$\frac{1}{T} \sum_{t=1}^{T} p_t \to \mathbb{E}_\pi(p)$$

Therefore, the burn-in period were selected conservatively using the results for $L = 50$, the largest system considered. The first 10000 and 5000 observations were discarded for the BTW and Manna model respectively. Simulations were run for 60000 and 55000 grain drops respectively, leaving 50000 observations post burn-in for every model and system sizes. While the stability of the mean height supports these burn-in choices, we may note that it does not by itself prove that every aspect of the system has reached stationarity.

3.1.1 Implementation

The simulations and statistical analyses were implemented in Python and are provided in the submitted Jupyter notebook. Each sand pile was represented by an $L \times L$ integer-valued NumPy array. NumPy was used for numerical operations and pseudorandom number generation, Sklearn for computing R^2, Pandas for correlation, and Matplotlib for visualisation. A fixed random seed `12345` was used to make the simulations reproducible.

To perform an avalanche, an initial implementation searched the full lattice for unstable sites. However, this $O(L^2)$ algorithm became very computational expensive especially for larger L values. Hence this was later replaced using a first-in, first-out queue with Python's `collections.deque`, which supports append and removal

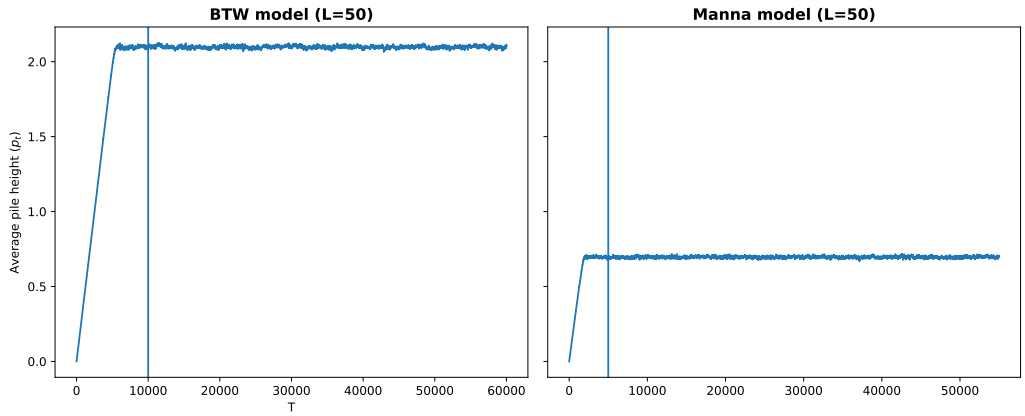


Figure 3.1: Mean pile height at each time step for the BTW and Manna models with $L = 50$. The vertical lines indicate the burn-in periods used in the subsequent analysis.

operations from either end in approximately constant time [13]. Only sites whose heights could have changed were added to the queue, avoiding repeated full-lattice searches and improving the efficiency of the simulation.

3.2 Avalanche properties

In this report, we investigate four avalanche properties:

1. **Topples**. The number of topples required to take the system back to a stable configuration;
2. **Area.** The number of unique sites that have been toppled;
3. **Loss.** The number of grains dissipate off the open boundaries;
4. **Length.** The maximum L_1-distance from the avalanche origin (x_0, y_0) where the sand is dropped, to any toppled sites.

$$L = \max_{(x,y)\in A} \{|x_0 - x| + |y_0 - y|\}$$

where A are the sites that has been toppled.

L_1 distance gives us an intuitive interpretation of length as it measures the minimum number of lattice steps between an avalanche origin and a toppled site (since we only move in vertical or horizontal directions). Note that this simpler definition differs from the "radius of gyration" previously discussed in Chapter 1.

Positive associations are expected between topples, area, length and possibly loss, since an avalanche having more topplings is more likely to affect more unique sites and propagate further from its origin. Associations with loss may be weaker since dissipation only happens when an avalanche reaches the open boundary. To examine this, we calculate pairwise Spearman rank correlations between all four properties. Spearman correlation [14] was selected because it measures monotonic association and is less sensitive than Pearson correlation to the extreme observations produced by heavy-tailed distributions.

3.3 Empirical distribution

Many sand drops produce no avalanches. Power-law analysis is therefore restricted to positive observations since $\log x$ is undefined at $x = 0$. However, we did not exclude zero values simply due to this reason. For topples and area, a zero value indicates that no avalanche occurred, hence it makes natural sense to examine its positive distribution. Meanwhile, a zero length may also represent an avalanche confined to its origin, and a zero loss indicates no boundary dissipation whether or not an avalanche has happened. Hence, we report $P(X = 0)$ separately for loss and length.

For visual clarity when comparing across different system sizes, the empirical distributions were displayed using 30 logarithmically spaced bins. Log binning was used only for visualisation and the cutoff selection and fitted exponents were performed from other methods. This prevents the estimated exponents from depending on the selected number of bins.

3.4 Power law fitting

Throughout this report, log denotes the natural log. To identify an approximate power-law region, we first calculate the survival function $S(x) = P(X < x)$. This is smoother than our empirical PMF, making the departure from an approximately straight region easier to identify . For each candidate upper cutoff, a straight line was fitted to $\log S(x)$ against $\log x$ using observations up to that cutoff. The largest candidate satisfying the chosen R^2 threshold was selected. We used $R^2 \geq 0.95$ for topples and area and $R^2 \geq 0.88$ for length. The threshold was lowered for length since no for practical purposes and will be later discussed in Chapter 5. These thresholds only serves as practical criteria for identifying approximately linear regions of a power-law distribution.

Theoretically, for an untruncated power law, we have,

$$P(x) \propto x^{-\tau}$$

And the survival function should therefore asymptotically satisfy

$$S(x) = P(X \geq x) \propto x^{1-\tau}$$

Nevertheless, in our finite system, the survival function may deviate from this ideal asymptotic relationship due to discreteness of observables x, their finite and relatively small range, as well as the upper tail truncation. Therefore the survival function slope was used only to select the cutoff and was not reported as the final exponent.

After cutoff selection, a common exponent was estimated directly from the unbinned empirical PMF values within the selected scaling regions. Values from all system sizes were pooled and fitted by ordinary least squares (OLS) in log-log space:

$$\log P(x) = -\tau \log(x) + C$$

This produced a common exponent τ describing the family of finite size distributions. For the Manna model, values below the first non-plateau log bin were excluded as a lower cutoff. Nevertheless, note that the estimated exponent remains sensitive to the selected upper cutoff. If the cutoff extends into the sparse finite-size tail, tail observations can influence the fitted slope. This limitation will be discussed further in Chapter 5.

To investigate the log finite size correction as proposed by Yadav et al. [3], the probability distributions are rescaled by $\log N$ where $N = L^2$. Hence the rescaled response variable becomes

$$\log(\log(N)P(x)) = \log P(x) + \log\log N$$

A new common fit was then calculated, τ':

$$\log(\log(N)P(x)) = -\tau' \log(x) + C'$$

The quality of the common collapse was compared before and after rescaling using the overall RMSE and maximum residual in log scale. Smaller values indicate that the curves agree closer to their common fitted line.

3.5 Exponential fitting

Exponential distributions were considered for the loss and length. $P(X = 0)$ is reported separately for both properties and a log-linear model is fitted to the empirical probabilites for $x > 0$ values.

$$\log P(x) = -\lambda x + C$$

Here, λ corresponds to the decay rate. The larger the value, the faster the decay.

We use R^2 value and RMSE values to examine the fit and use residual plots for diagnosing systematic patterns. In particular, curvature or an S-shaped residual pattern indicates that a single exponential decay rate may not adequately describe the observed distribution.

Chapter 4

Results

This chapter presents the numerical results for the distributions of topples, area, loss, and length. The selected upper cutoffs for the power-law analysis are reported in Table 4.1. For the Manna topples and area distributions, a fixed lower cutoff of $x_{\min} = 2$ was used for every system size, excluding the $x = 1$ observations associated with the initial plateau. All other power-law fits used $x_{\min} = 1$.

Property	Model	$L = 10$	$L = 20$	$L = 30$	$L = 40$	$L = 50$
Topples	BTW	29	90	212	339	540
	Manna	47	261	870	1947	3288
Area	BTW	27	78	177	269	424
	Manna	23	100	246	478	721
Length	BTW	8	15	24	32	40
	Manna	8	18	28	39	50

Table 4.1: Selected upper cutoffs for the power-law fits. A fixed lower cutoff of $x_{min} = 2$ was used for the Manna topples and area distributions; all other fits used $x_{min} = 1$

4.1 Topples

Figure 4.1 shows the distributions of the number of topples for the BTW and Manna models, before and after $\log N$ scaling. For both models, the distributions exhibit an approximately straight intermediate region on log-log axes, followed by a system size dependent tail. The tail extends to larger number of topples as L increases, indicating a finite-size cutoff.

Model	$\log N$ scaling	τ	Overall RMSE
BTW	before	1.120	0.311
	after	1.070	0.299
Manna	before	1.123	0.478
	after	1.084	0.491

Table 4.2: Common power-law exponent estimates and overall RMSEs (in log scale) for the topples distributions.

For the BTW model, the common fit before $\log N$ scaling is $\tau_s = 1.120$, compared to $\tau_s' = 1.070$ after the scaling. The overall RMSE decreases from 0.311 to 0.299, corresponding to a slight reduction. For the Manna model, the corresponding exponents are 1.123 and 1.084 respectively, which lie very close to the BTW estimates. In contrast, the overall RMSE changes from 0.478 to 0.491, a slight increase.

Since the overall RMSE is calculated based on pooled data, we may have a closer look on the maximum deviations across each L in Table 4.3. For the BTW model, the maximum deviations decreased substantially for $L = 10$ and $L = 20$, remained approximately unchanged for $L = 30, 40, 50$. The largest deviation in both cases occurred for $L = 50$. Hence it seems after the system size dependence adjustment proposed by Yadav et al. [3], the smaller grid sizes agree more with the collapse.

For the Manna model, the maximum deviations showed no uniform response to the adjustment. They decreased for $L = 10$, $L = 40$ and $L = 50$, but increased for $L = 20$ and $L = 30$. The Manna distributions also

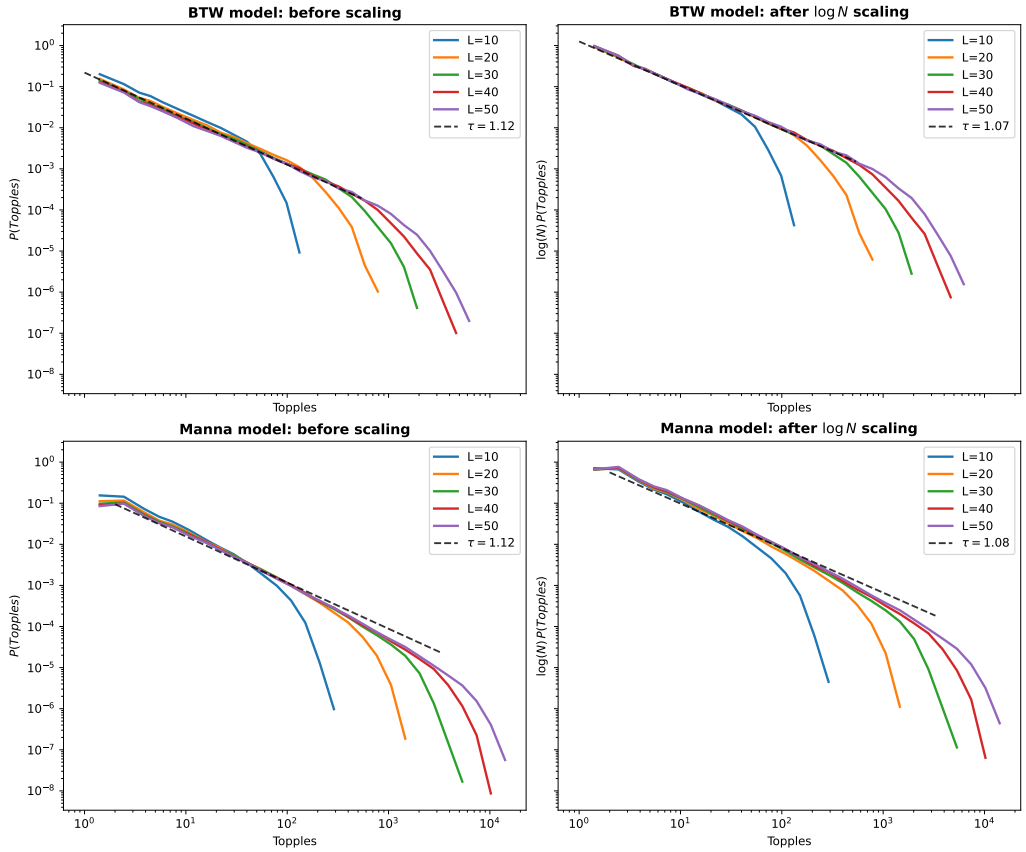


Figure 4.1: Log-binned topples distributions for the BTW model (top row) and Manna model (bottom row), before scaling (left column) and after $\log N$ scaling (right column). Dashed lines show the common power-law fits.

Model	$\log N$ scaling	$L = 10$	$L = 20$	$L = 30$	$L = 40$	$L = 50$
BTW	before	0.449	0.504	0.609	1.068	1.900
	after	0.310	0.327	0.610	1.080	1.879
Manna	before	0.524	1.487	1.765	1.551	1.818
	after	0.359	1.662	1.829	1.541	1.740

Table 4.3: Maximum absolute log-space residuals for the topples distributions. The adjustment reduces deviations for the smaller BTW systems, while the Manna results show no uniform improvement.

displayed a short plateau at small numbers of topples before entering the approximately linear region. This plateau was excluded using the lower cutoff described earlier.

Overall, the $\log N$ adjustment produced a modest improvement for the BTW model, as indicated by the overall RMSE and the reductions in maximum deviations for small L values. No consistent improvement was observed for the Manna model, since the overall RMSE increased and its maximum deviations displayed a mixed pattern across system sizes.

4.2 Area

Figure 4.2 shows the area distributions for BTW and Manna models. The distributions displayed very similar finite size behaviour as we observed in the topples section, an approximately straight region followed by a system size dependent tail.

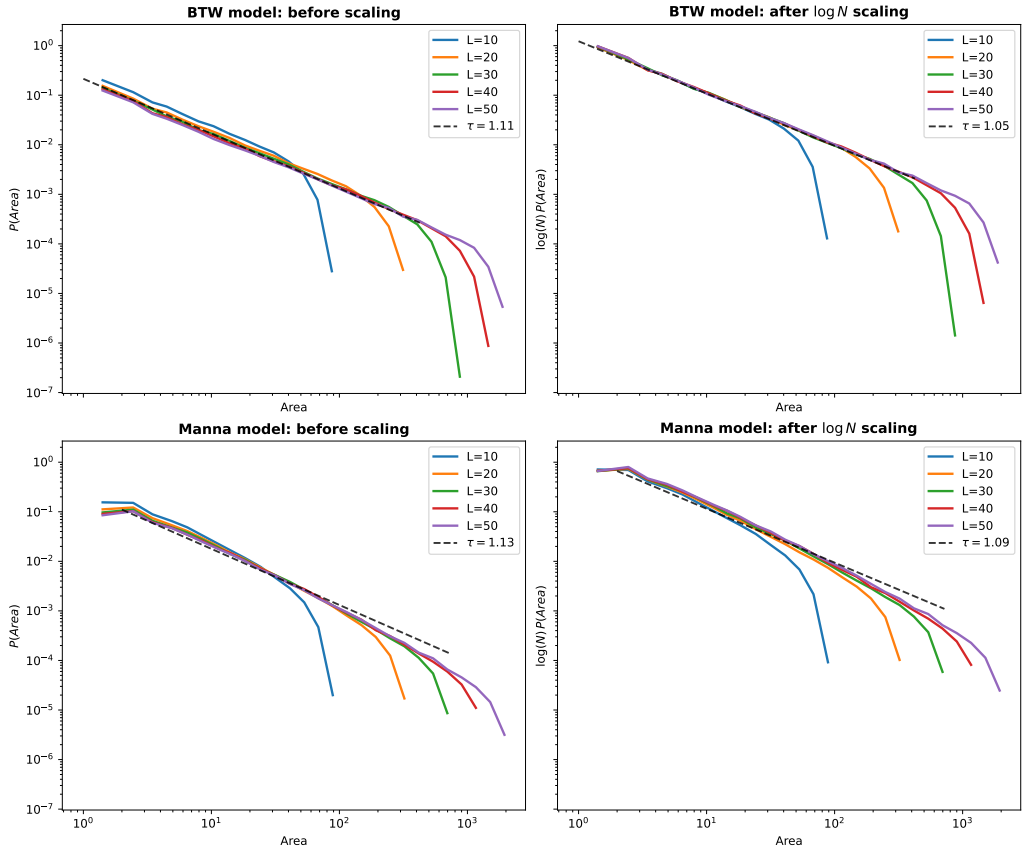


Figure 4.2: Log-binned area distributions for the BTW model (top row) and Manna model (bottom row), before scaling (left column) and after $\log N$ scaling (right column).

As shown in Tables 4.4 and 4.5, for the BTW model, the area exponent changed from $\tau_a = 1.106$ to $\tau_a' = 1.054$ after the $\log N$ scaling. The overall RMSE decreased from 0.270 to 0.256, and maximum deviations showed a very similar pattern as in the topples case (substantially decreased for $L = 10$ and $L = 20$, approximately unchanged for $L = 30, 40, 50$). This may indicate after the rescaling, results from smaller grid sizes show more agreement with the common collapse.

Model	$\log N$ scaling	τ	Overall RMSE
BTW	before	1.106	0.270
	after	1.054	0.256
Manna	before	1.133	0.365
	after	1.088	0.358

Table 4.4: Common power-law exponent estimates and overall RMSEs (in log scale) for the area distributions.

For the Manna model, the corresponding exponent estimated decreased from 1.133 to 1.088, RMSE from

0.365 to 0.358. In constrast to the topple case, the maximum deviations mainly remained approximately unchanged or decreased substantially for smaller L values.

Model	$\log N$ scaling	$L = 10$	$L = 20$	$L = 30$	$L = 40$	$L = 50$
BTW	before	0.493	0.433	0.659	0.897	1.263
	after	0.283	0.282	0.698	0.910	1.252
Manna	before	0.324	0.514	0.956	1.764	1.670
	after	0.258	0.337	1.047	1.797	1.649

Table 4.5: Maximum absolute log-space residuals for the area distributions.

Therefore, the effect of the $\log N$ adjustment on the area distributions differs for the topple distributions, as now for both BTW and Manna models, it modestly improved the collapse for smaller system sizes.

4.3 Loss

Table 4.6 reports the probability of zero loss and exponential fitting results. For both models, $P(loss = 0)$ increased with L. This is reasonable is since the number of boundary sites only increases proportionally with L while our grid size increases with L^2. Avalanches in larger systems are therefore less likely to reach an open boundary.

The fitted decay rate λ generally decreased with L, meaning that the positive loss distribution decayed slower for larger systems. The Manna model generally produced smaller zero loss probabilities than the BTW model. This indicates that boundary dissipation occurred more frequently under its stochastic toppling rule.

Model	Statistics	$L = 10$	$L = 20$	$L = 30$	$L = 40$	$L = 50$
BTW	$P(loss = 0)$	0.723	0.766	0.793	0.816	0.826
	λ	0.327	0.227	0.163	0.137	0.137
	R^2	0.992	0.953	0.936	0.905	0.929
Manna	$P(loss = 0)$	0.589	0.676	0.724	0.758	0.785
	λ	0.474	0.252	0.199	0.145	0.118
	R^2	0.994	0.939	0.952	0.930	0.902

Table 4.6: Zero-loss probabilities, exponential decay rates, and R^2 values for the loss distributions.

Figure 4.3 shows the log probabilities of positive loss together with the fitted exponential models. The approximately linear trends on log-normal axes produced relatively high R^2 values, ranging approximately from 0.90 to 0.99.

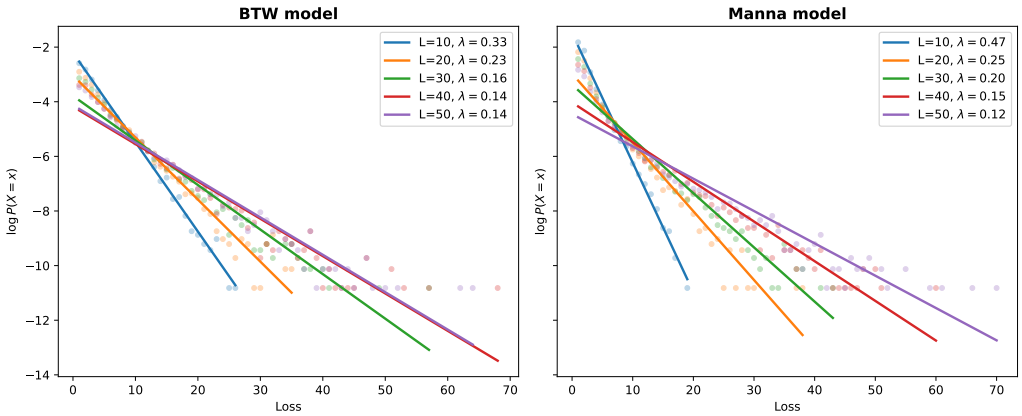


Figure 4.3: Positive loss distributions for BTW model (left) and Manna model (right), with fitted exponential models.

Nevertheless, Figure 4.4 shows systematic U-shaped residual patterns for most system sizes. Residuals were generally positive for small losses, negative over the intermediate range, and positive again in the far right tail. Thus, the exponential model tends to underestimate the probabilities of small and very large losses while overestimating intermediate probabilities. Therefore, for our limited selected grid sizes L, it seems an exponential model may not fully capture the shape of the positive-loss distribution.

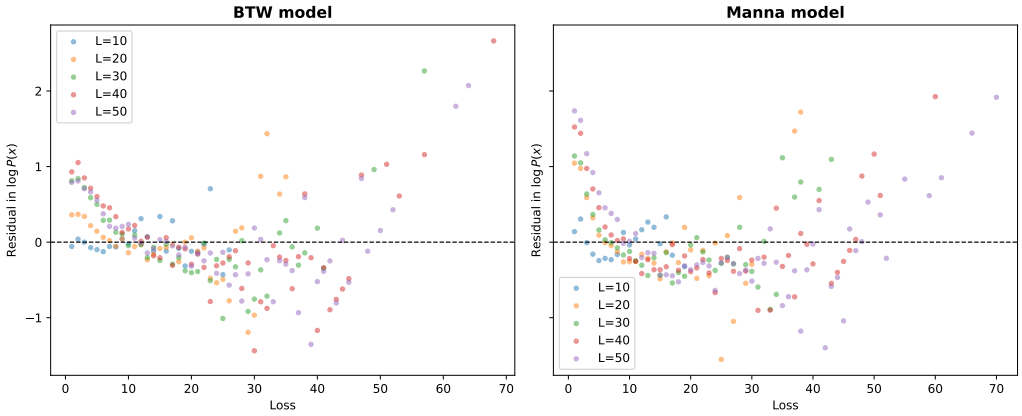


Figure 4.4: Residuals in $\log P(x)$ from the exponential fits for BTW model (left) and Manna model (right). The dashed horizontal line denotes zero residual.

4.4 Length

4.4.1 Power law

As mentioned in Chapter 3, we lowered the R^2 threshold for length because the higher threshold did not identify with a broad enough fitting region. Figure 4.5 clearly illustrates more visible curvatures within the candidate regions than what we observed for area and topples.

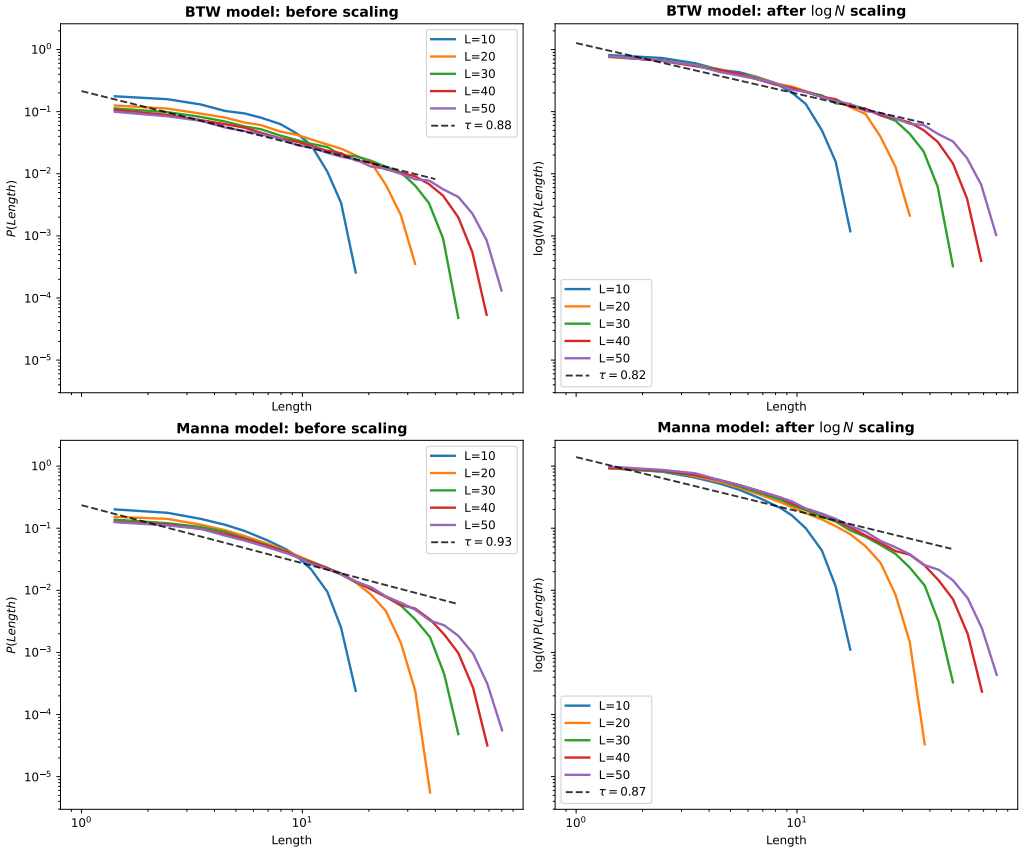


Figure 4.5: Log-binned length distributions for the BTW model (top row) and Manna model (bottom row), before scaling (left column) and after $\log N$ scaling (right column).

Table 4.7 shows that for BTW model, the estimated exponent changed from $\tau_l = 0.884$ to $\tau_l' = 0.816$ after $\log N$ scaling. The RMSE decreased from 0.223 to 0.152. We may observe a similar pattern in the Manna model, exponent estimates are 0.934 and 0.869 respectively, larger than estimates in BTW model. The RMSE decreased from 0.228 to 0.171. The change is more substantial than the previous cases with area and topples.

Model	$\log N$ scaling	τ	Overall RMSE
BTW	before	0.884	0.223
	after	0.816	0.152
Manna	before	0.934	0.228
	after	0.869	0.171

Table 4.7: Common power-law exponent estimates and overall RMSEs (in log scale) for length.

Notably, in both models, maximum deviations in Table 4.8 show consistent and generally substantial decrease across all L values after $\log N$ scaling. Therefore, although the length distributions showed weaker survival function linearity than the topple and area distributions, the common power law fit produced relatively small residuals, and the $\log N$ adjustment substantially reduced the both the overall RMSE and maximum deviations across L. This indicates improved alignment between system sizes within the selected scaling regions.

Model	$\log N$ scaling	$L = 10$	$L = 20$	$L = 30$	$L = 40$	$L = 50$
BTW	before	0.600	0.532	0.640	0.701	0.766
	after	0.446	0.524	0.505	0.484	0.490
Manna	before	0.603	0.624	0.732	0.793	0.858
	after	0.539	0.616	0.597	0.576	0.583

Table 4.8: Maximum absolute log-space residuals for length.

4.4.2 Exponential

An exponential model was also additionally examined for length because the unscaled distributions displayed visible curvature. However, its residuals exhibited a systematic S-shaped pattern, indicating that a single exponential decay rate did not describe the complete positive-length distribution.

Model	Statistics	$L = 10$	$L = 20$	$L = 30$	$L = 40$	$L = 50$
BTW	$P(loss = 0)$	0.689	0.651	0.633	0.628	0.621
	λ	0.417	0.173	0.127	0.093	0.068
	R^2	0.863	0.914	0.889	0.910	0.920
Manna	$P(loss = 0)$	0.448	0.399	0.381	0.369	0.365
	λ	0.390	0.219	0.138	0.104	0.085
	R^2	0.904	0.909	0.951	0.952	0.960

Table 4.9: Zero-length probabilities, exponential decay rates, and R^2 values for the length distributions.

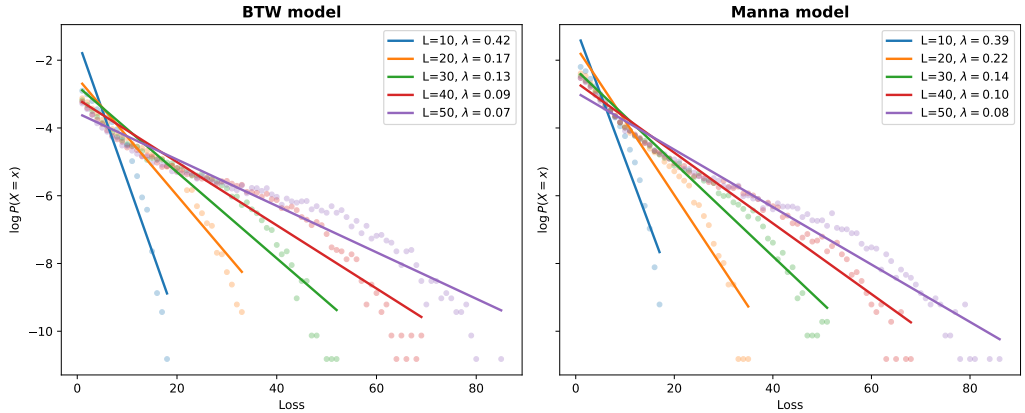


Figure 4.6: Positive length for the BTW model (left) and Manna model (right), with fitted exponential models.

As shown in Figure 4.6 and Table 4.9, the zero loss probability and decay rate both decrease as L increases. This means although zero loss cases became less common, the rate of decay also decreases. This can suggest a

broader distribution of positive length.

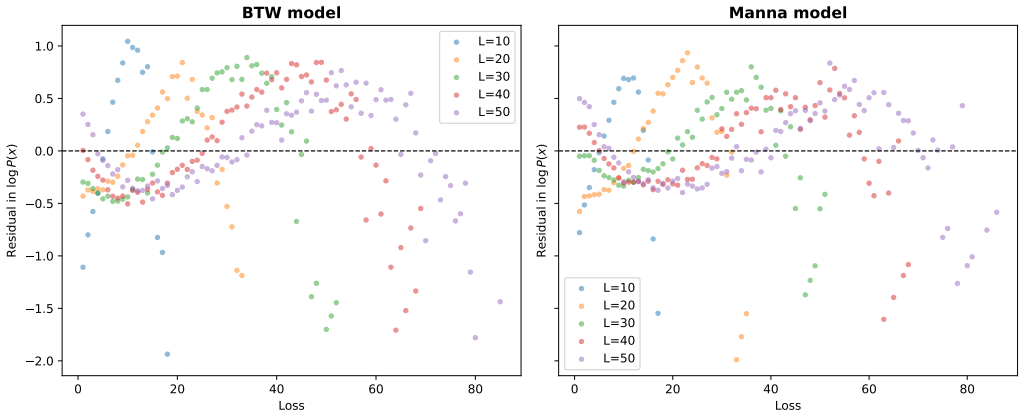


Figure 4.7: Residuals in $\log P(x)$ from the exponential length fits for BTW model (left) and Manna model (right). The systematic S-shaped patterns indicate misfit of a single exponential model.

The diagnostic residual plot in Figure 4.7 shows a clear S-shaped pattern for most L values. This indicates a systematic misfit of model. That is, a single exponential model does not adequately capture the positive length distribution. We may observe the same curvature in Figure 4.6 as scattered around the fitted lines.

Overall the results may favour an approximate power-law description with FSS, however the evidence is weaker than for topples and area. Neither model fully describes the positive length distribution.

4.5 Correlations

The pairwise correlations between avalanche observables were measured using Spearman rank correlation. As previously discussed, Spearman correlation was preferred because the data were discrete and strongly right-skewed. The correlations matrix shows small variations in value across L values, we only display the result for $L = 50$ here and the rest is left in Appendix A.

	Topples	Area	Loss	Length
Topples	1.000	-	-	-
Area	1.000	1.000	-	-
Loss	0.626	0.626	1.000	-
Length	0.963	0.964	0.625	1.000

Table 4.10: Pairwise Spearman's correlation for the BTW model, $L = 50$

	Topples	Area	Loss	Length
Topples	1.000	-	-	-
Area	0.999	1.000	-	-
Loss	0.527	0.527	1.000	-
Length	0.982	0.984	0.516	1.000

Table 4.11: Pairwise Spearman's correlation for the Manna model, $L = 50$

The result is consistent as what we have discussed previously in Section 3.2. Topples, area and length are strongly and positively correlated for both models. Correlations involving loss were positive but noticeably weaker, since loss depends specifically on whether an avalanche reaches the open boundary.

Chapter 5

Discussion

5.1 Power law behaviour

Power-law behaviours are theoretically plausible for topples, area and length. This is because an SOC state has no intrinsic characteristic avalanche scale, as we previously discussed in Chapter 2. The distribution therefore behaves approximately as a power law within an intermediate scaling region that is truncated by the finite lattice. In particular, length measures the distance of an avalanche travels, area measures the number of distinct sites within the affected region, and topples measure the total "activity" generated during the process that take the system back to a stable configuration. Hence if an avalanche remains scale invariant, all these closely related properties should remain scale invariant and are hence expected to exhibit approximately scale-free distributions. This argument does not apply as directly to loss, because loss depends specifically on whether an avalanche reaches the open boundary, which we will discuss in detail under Section 5.3.

From our simulation results, the observed topples and area distributions are consistent with this general picture. Figures 4.1 and 4.2 illustrates an approximate linear segment on log-log axes, followed by system size dependent tails. This indicates that avalanches occur over a wide range of scales while the largest observed topples and area are restricted by the finite lattice.

For the BTW model, the unscaled estimates $\tau_s = 1.120$ and $\tau_a = 1.106$ are lower than many of the literature estimates at around 1.3. The corresponding Manna estimates $\tau_s = 1.123$ and $\tau_a = 1.133$ are also lower than previously reported values [2]. Several methodology differences may contribute to this discrepancy, including our much smaller lattice sizes, the selected fitting cutoff intervals, and differences in exponent-estimation methods.

The interpretation of length is more difficult. The unscaled estimates were $\tau_l = 0.884$ for BTW and $\tau_l = 0.934$ for Manna. But since our maximum Manhattan distance is not equivalent to the radius of gyration studied by Lübeck and Usadel, our τ_l is not directly comparable with their reported τ_r. Moreover, the visible curvature and the lower threshold ($R^2 \geq 0.88$) required for length indicate weaker evidence for a power-law region. Hence, these estimates may not accurately reflect the asymptotic length exponent.

In addition, the Manna topples and area distributions exhibited an initial plateau. This can reflect discreteness or that very small local avalanches follow a different distribution. Excluding $x = 1$ produced only a small change in the estimated exponents (e.g., 1.123 to 1.121 for topples and 1.133 to 1.127 for area, unscaled), suggesting limited effect to this particular lower-cutoff choice. The selected upper cutoffs, however, were more sensitive to the chosen R^2 threshold and could extend into the sparse finite-size tail.

Overall, for topple, area and length, the intermediate power-law like regions and the extension of the cutoffs with L (as shown in Table 4.1) are consistent with scale-invariant avalanche behaviour and hence with SOC. Nevertheless, the evidence could be weaker for length and exponent estimates for Manna topples and area may have been affected by cutoff selections. Stronger evidence would require larger lattices, wider scaling regions, uncertainty estimates, and evidence that the fitted exponents stabilise as L increases.

5.2 Effect of system-size dependent scaling

In this report, we tested system-size dependent scaling proposed by Yadav et al [3] to topples, area and length. We wanted to investigate if this will lead to an improvement in collapse. That is, do the distributions for different system sizes lie closer to a common fitted line? We assessed this using two measures, an overall RMSE

and maximum absolute deviations across all L. They are complementary with each other as RMSE assesses the overall agreement while the max deviations identify which system sizes improves or deteriorates.

For the BTW topples and area distributions, the $\log N$ adjustment produced modest improvements. The overall RMSE decreased by approximately 3.9% for topples and 5.2% for area, with the largest reductions in maximum deviation occurring for $L = 10$ and $L = 20$. The adjusted exponents were also closer to the values near one proposed by Yadav et al. [3]. These findings are consistent with a logarithmic system-size dependence, although the small RMSE reductions and limited range of L cannot provide conclusive evidence.

The similar behaviour of topples and area is reasonable because avalanches affecting more distinct sites generally generate more toppling events. Their close relationship is also reflected by the near-perfect Spearman correlations observed for both models.

We additionally applied the $\log N$ adjustment to length, which was not considered in the original analysis by Yadav et al. The result is rather surprising as for BTW, the overall RMSE decreased massively by approximately 32%, and the maximum deviation decreased for every system size. This was the largest relative improvement among BTW observables. Nevertheless, improved alignment does not by itself establish that length follows a power law, especially because our original length distributions exhibited visible curvature.

The Manna results were less uniform. For topples, the overall RMSE increased by approximately 2.7%, and the maximum deviations showed a mixture of increases and decreases. There is therefore no consistent evidence that the adjustment improves the Manna topples collapse. For area, the RMSE decreased by approximately 1.9%, with most of the improvement again occurring for the smaller system sizes. This provides weak evidence of improved alignment.

For Manna length, the overall RMSE decreased by approximately 25% and the maximum deviations decreased for every system size. As in the BTW case, this suggests that the adjustment reduces vertical separation between the length distributions. However, because the underlying power-law evidence is weaker, we may only take this improvement as an improved empirical alignment.

Overall, the $\log N$ adjustment is most consistently supported by BTW area and topples and by the empirical alignment of length. Its effect on the Manna model can depend on avalanche property, since result is inconclusive for Manna topple, weak supportive evidence for Manna area and improved empirical alignment for Manna length.

5.3 Exponential behaviour

Unlike topples, area and length, loss only happens when an avalanche reaches an open boundary. It is therefore not as directly related to the scale-free characteristics of an avalanche.

The loss results provide partial support for an exponential behaviour. The positive-loss probabilities appear approximately linear on log-normal axes, and the corresponding R^2 values are relatively high. Moreover, $P(X = 0)$ increases with system size for both models. This is consistent with the decreasing ratio of boundary sites to total sites as L increases, since total sites increases proportionally to L^2 while boundary site only increases proportionally to L. Therefore, a randomly initiated avalanche must propagate further on average to reach an open boundary. The Manna model has a lower zero-loss probability than the BTW model at every system size, indicating that boundary dissipation occurs more frequently in the simulated Manna avalanches.

Nevertheless, the residual plots show a systematic U-shaped pattern. The fitted model generally underestimates the probabilities of small and very large losses while overestimating probabilities in the intermediate range. Therefore, the relatively high R^2 values should not be seen as evidence of an exact exponential distribution. A single decay rate λ hence cannot fully describe its shape.

An exponential model was also tested for avalanche length because the unscaled length distributions showed noticeable curvature and weaker power-law linearity than the topples and area distributions. Although some of the log-normal fits produced moderately high R^2 values, their residuals exhibited a clear S-shaped pattern. This systematic curvature shows that a single exponential distribution is not an adequate model for positive length.

The failure of the exponential model for length is actually also consistent with its earlier power-law interpretation. Length measures the distance travelled by an avalanche and is therefore more directly connected to

scale-invariant avalanche property than loss. This can be supported by decreasing zero-length probabilities and decay rate as L increase, which may indicate a broader range of length values.

Therefore, the results favour an approximate power-law description for length within a restricted finite-size scaling region over a single exponential description of the complete distribution. However, the visible curvature and the lower cutoff-selection threshold still mean that the evidence for power-law behaviour in length remains weaker than for area and topples

5.4 BTW vs. Manna model

Despite their different toppling rules, both models displayed several common features. Their topples and area distributions contained approximately power-law intermediate regions followed by finite-size cutoffs, and their fitted exponents were relatively close to each other across the system sizes we considered. Both models also exhibited strong positive correlations among topples, area, and length.

Nevertheless, several differences were also observed. The Manna distributions exhibited an initial plateau that was not present or at least as prominent in the BTW results. Notably, the $\log N$ adjustment did not consistently improve the Manna topples collapse. The Manna model also produced a lower probability of zero loss, suggesting that its stochastic toppling rule generated avalanches that reach boundaries more frequently. These differences indicate that the toppling rule can affect of the finite-size distributions for some avalanche properties.

5.5 Limitations

The main computational limitation of this exploration is the restricted range of system sizes. Simulations were limited to $L = 50$, whereas literature exponent estimates usually used substantially larger lattices. Therefore the fitted slopes may not represent their asymptotic values. Attempts were made to resolve this computational issue. We used a queue-based implementation to replace the initial whole grid-search strategy and improved the toppling efficiency. Nevertheless, the computational cost of the Manna model restricted further increases in L since we wanted our simulation results to be easily reproducible with a relatively short amount of time.

The exponent estimates are also sensitive to the selected fitting regions. Upper cutoffs were chosen using an R^2 threshold applied to the empirical survival function, but this could allow some selected scaling regions to extend into the sparse finite-size tail. Furthermore, ordinary least squares was applied to unbinned log-PMF points, giving each distinct observed value equal weight even when tail values occurred rarely, hence potentially biasing the exponent estimates. Alternative cutoff thresholds or likelihood-based power-law methods could therefore produce different estimates.

Finally, each result was based on one reproducible simulated trajectory for each model and system size. Although 50000 post-burn-in observations were retained, repeated simulations with independent seeds would be needed to quantify uncertainty intervals. Our collapse analysis also measured agreement with a common fitted line after vertical $\log N$ rescaling rather than performing a complete finite-size collapse involving estimation of the horizontal scaling dimension D_x. In addition, our Manhattan-distance definition of length is not directly comparable with the radius of gyration commonly used in the literature.

Appendix A

Correlations

	Topples	Area	Loss	Length
Topples	1.000	-	-	-
Area	1.000	1.000	-	-
Loss	0.643	0.644	1.000	-
Length	0.976	0.979	0.627	1.000

Table A.1: Pairwise Spearman's correlation for the BTW model, $L = 10$

	Topples	Area	Loss	Length
Topples	1.000	-	-	-
Area	0.994	1.000	-	-
Loss	0.482	0.483	1.000	-
Length	0.961	0.973	0.461	1.000

Table A.2: Pairwise Spearman's correlation for the Manna model, $L = 10$

	Topples	Area	Loss	Length
Topples	1.000	-	-	-
Area	0.999	1.000	-	-
Loss	0.575	0.575	1.000	-
Length	0.981	0.984	0.560	1.000

Table A.3: Pairwise Spearman's correlation for the BTW model, $L = 20$

	Topples	Area	Loss	Length
Topples	1.000	-	-	-
Area	0.996	1.000	-	-
Loss	0.474	0.474	1.000	-
Length	0.970	0.979	0.460	1.000

Table A.4: Pairwise Spearman's correlation for the Manna model, $L = 20$

	Topples	Area	Loss	Length
Topples	1.000	-	-	-
Area	0.999	1.000	-	-
Loss	0.558	0.558	1.000	-
Length	0.984	0.987	0.545	1.000

Table A.5: Pairwise Spearman's correlation for the BTW model, $L = 30$

	Topples	Area	Loss	Length
Topples	1.000	-	-	-
Area	0.997	1.000	-	-
Loss	0.473	0.472	1.000	-
Length	0.973	0.981	0.459	1.000

Table A.6: Pairwise Spearman's correlation for the Manna model, $L = 30$

	Topples	Area	Loss	Length
Topples	1.000	-	-	-
Area	0.999	1.000	-	-
Loss	0.575	0.575	1.000	-
Length	0.981	0.984	0.560	1.000

Table A.7: Pairwise Spearman's correlation for the BTW model, $L = 40$

	Topples	Area	Loss	Length
Topples	1.000	-	-	-
Area	0.996	1.000	-	-
Loss	0.474	0.474	1.000	-
Length	0.970	0.979	0.460	1.000

Table A.8: Pairwise Spearman's correlation for the Manna model, $L = 40$

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